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# No Association Between Fencing Response After Concussion and Recovery Time, Concussion Severity, or Clinical Outcomes in Professional American Football Players

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**BACKGROUND AND OBJECTIVES:** The relationship between tonic posturing (also known as the "fencing response") after concussion and clinical outcomes is unclear. The objective of this study was to compare clinical outcomes between professional American football players with a documented fencing response after concussion to players without documented fencing response after concussion.

**METHODS:** Players with a fencing response after concussion were matched 1:2 to players without documented fencing response after in-game concussion by age, position, and concussion history from the 2018 to 2023 seasons. The primary outcome measure was number of days until return to full participation. Secondary outcome measures included screening of neurocognitive function with the Standardized Assessment of Concussion (SAC) and modified Balance Error Scoring System (mBESS), and subjective symptom reporting with the Sport Concussion Assessment Tool-5. An independent-samples median test compared game-day SAC and mBESS outcomes between groups. A Kaplan-Meier curve evaluated fencing response after concussion on days until return to full participation. A linear mixed-effects model for 11 Sport Concussion Assessment Tool-5 total symptom scores (ie, day of injury up to postinjury day 10) compared symptom recovery over time between fencing groups.

**RESULTS:** Fifty-seven fencing responses were documented over 6 seasons. Nearly 2-in-3 players with a fencing response played defensive back or wide receiver. There was no difference between players with a fencing response (median = 10 days) and those without fencing response (median = 9 days) in days until return to full participation ($P$ = .16). Fencing response was not associated with a change in hazard for days until return to full participation ($\chi^2$ = 0.91; $P$ = .34). There was no difference between groups in SAC scores ($P$ = .18-.99) or mBESS score (fencing: 4, nonfencing: 4; $P$ = .87). There was no difference between groups in symptom scores over time (β: −0.02; $P$ = .92).

**CONCLUSION:** Fencing response after concussion was not associated with symptom recovery, concussion severity, or short-term recovery in professional American football.

**KEY WORDS:** Fencing response, Tonic posturing, Concussion, Professional football, Clinical outcomes

Sport-related concussion (SRC) remains a public health concern in the United States, with a broad public discourse about potential long-term effects of concussion and/or repetitive head impacts.[1,2] Concussion evaluation and diagnosis is largely reliant on the presence of transient neurological signs of injury, subjective symptoms, and standardized clinical assessments because neuroimaging modalities such as computed tomography scans and MRI rarely show intracranial pathology (approximately 1%-5% of

**ABBREVIATIONS: EMR,** electronic medical record; **mBESS,** modified Balance Error Scoring System; **NFL,** National Football League; **NFLPA,** NFL Players Association; **SAC,** Standardized Assessment of Concussion; **SCAT-5,** Sport Concussion Assessment Tool-5; **SRC,** sport-related concussion; **UNC,** Unaffiliated Neurotrauma Consultant.

concussion patients).[3,4] Indeed, recent diagnostic criteria for concussion excludes intracranial pathology in the definition.[5,6] As a result, concussion has historically been considered a less severe head injury because of the lack of structural abnormalities on traditional imaging modalities and its shorter recovery time.[7]

Certain signs of concussion, such as loss of consciousness, were initially considered to be indicative of concussion severity.[8] More recent data indicate loss of consciousness may not be as impactful on recovery time as initially suspected, though data are mixed.[9,10] Another concussion sign is tonic posturing, colloquially referred to as the "fencing response,"[11] which is a reflex of the central nervous system characterized by extension of 1 upper extremity and flexion of the contralateral upper extremity with an altered sensorium after head trauma.[12] Observing the fencing response can be unsettling, which may contribute to the public perception that a concussion with a fencing response may be a more severe brain injury. The fencing response mimics the asymmetrical tonic neck reflex observed naturally in infants, wherein ipsilateral extension and contralateral flexion of the upper extremities is elicited when an infant's head is turned to the side.[13] The asymmetrical tonic neck reflex is thought to aid in developing muscle tone and hand-eye coordination in infancy.[14] The reflex is mediated by the lateral vestibular nucleus which modulates postural adjustments based on afferent feedback during movement.[15] The asymmetrical tonic neck reflex disappears gradually by approximately 6 months of age in infants,[14] but activation of the lateral vestibular nucleus through biomechanical rotational forces applied to the head of animals has been shown in animal models to elicit tonic posturing.[12]

Despite public awareness of the fencing response and discussion in popular media,[1] the relationship between fencing response and clinical recovery from concussion is not clear. The purpose of this study was to compare clinical outcomes between professional American football players with a documented fencing response to players without documented fencing response matched by age, position played, and concussion history. We hypothesized that there would be no difference between groups in clinical examination findings, symptom severity, or time until return to play because of the lack of available evidence on how fencing response alters concussion recovery.

## METHODS

Approval for this study was obtained from the University of Pittsburgh Institutional Review Board (#STUDY23060005; approved July 11th, 2023) and through the National Football League (NFL)-NFL Players Association (NFLPA) Player Scientific and Medical Research Approval Process, as outlined in the NFL-NFLPA Collective Bargaining Agreement.[16] Participant consent to potentially use their data for research purposes was obtained during player intake into the league.

### Design and Participants

This case-control study matched each professional American football player who exhibited a fencing response after concussion to 2 professional American football players without the fencing response after concussion. This matching design was selected to increase statistical power and account for concussion heterogeneity across individuals, both in presentation and recovery, and to account for player-factors (eg, age, position, concussion history). All participants had electronic medical record (EMR) documented in-game concussions with an unaffiliated neurotrauma consultant (UNC) and club physician evaluation. Data were obtained from the NFL injury analytics database from the 2018 to 2023 preseasons and regular seasons.[17] The UNCs documented characteristics of each injury in postevaluation reports. Observation of fencing response was documented in free-text comments. Any concussions documented as "questionable" or "probable" fencing response were excluded from analysis. Those who had the fencing response were matched to those without a fencing response based on age (within 1 year), number of prior concussions during the player's NFL career (sport related and nonsport related) documented in the NFL EMR (±1), and player position. Potential matches were ranked by closeness of match, with exact matches being selected first. If multiple control participants were matched with the same level of closeness, concussions occurring in the same year were matched first, followed by random selection. If 2 matches were not identified, match criteria were expanded to a larger age and concussion history window. For 2 players with a fencing response after concussion, only one control participant match could be identified. The final analyzed data set included 57 players with a fencing response after concussion and 112 players without a fencing response after concussion.

### Outcomes

The primary outcome measure of the study is median number of days from injury until return to full participation in football, as documented in the NFL EMR. Secondary outcome measures included both objective (Standardized Assessment of Concussion [SAC], modified Balance Error Scoring System [mBESS]) and subjective symptom assessments of concussion severity (Sport Concussion Assessment Tool-5 [SCAT-5]), documented in the C3Logix concussion management system. The SAC includes an objective cognitive examination assessing immediate memory recall (range: 0-30), orientation (range: 0-5), concentration (range: 0-5), and delayed memory recall (range: 0-10), with a possible total SAC score from 0-50. Higher scores indicate better performance.[18] Of note, all participants were administered the 10-item memory word list on the SAC. The mBESS is an objective assessment of postural stability in which players are asked to maintain 3 positions (ie, double leg stance, single leg stance, tandem stance) for 20 seconds with their eyes closed.[19] The number of deviations from the initial instructed position are counted as errors, for a maximum total score of 30 (where higher indicates worse postural stability). The SAC and mBESS were administered on the day of injury as part of the NFL-NFLPA game-day concussion protocol.[20] The symptom checklist from the SCAT-5 is a 22-item self-report symptom questionnaire.[21] SCAT-5 total symptom score ranges from 0 to 132, where higher scores indicate more symptoms. In the NFL population, SCAT-5 symptom scores are typically collected daily until full return to participation. SCAT-5 symptom data were analyzed daily from the day of injury until postinjury day 10.

### Statistical Analysis

The C3Logix system was implemented in 2019, so players injured in 2018 were included in analysis of the primary outcome measure but were missing data from the SAC, mBESS, and SCAT which excluded 23 participants from those analyses. Missing data from the remaining sample were observed for the SAC (13%) and SCAT5 (17%-69%) outcomes. Missingness was higher for daily SCAT symptom scores further out from injury because of variable recovery times in players (ie, symptoms were no

longer measured after the player returned to full participation). All outcome variables violated normality assumptions. Nonparametric independent samples median tests were used to compare players with fencing response after concussion to players without fencing response after concussion on days to return to full participation, SAC outcomes, mBESS total score, and SCAT5 symptom outcomes. Yates continuity corrections were applied to minimize the risk of Type I error. A linear mixed-effects model for the 11 SCAT5 total symptom scores was used to compare symptom recovery over time between players who exhibited a fencing response after concussion to players who did not exhibit a fencing response after concussion. Players were included in the linear mixed-effects model if they had at least 1 recorded total symptom score in the 11 day span. SCAT5 total symptom scores were log-transformed before analysis because of violating normality assumptions. A Kaplan-Meier survival analysis was used to test the equality of distributions for days until return to full participation between players with or without fencing response with log-rank Mantel-Cox comparisons. Alpha was set to $P < .05$. Analyses were conducted in the SPSS software (v29.0.2).

## RESULTS

Demographics for both groups are presented in Table 1. A total of 983 concussions occurred in preseason or regular-season games during the observational period, and 5.8% (n = 57) had a documented fencing response. Most concussions occurred during the regular season (83.4%). Over half of the cohort (~60%) had no prior concussion within their NFL career documented in the EMR. Nearly 2 in 3 players with a fencing response played defensive back or wide receiver.

### Association of Fencing Response With Days Until Return to Full Participation

Descriptive statistics for time until return to full participation between groups is provided in Table 2. There was no statistically significant difference between players with a fencing response (median = 10 days, range 4-155 days) and those without fencing response (median = 9 days, range 3-151 days) on time until return to full participation ($\chi^2 = 1.96$; $P = .16$). There was not a statistically significant difference in the survival distribution between those who exhibited the fencing response after concussion and those who did not regarding days until return to full participation ($\chi^2 = 0.91$; $P = .34$; Figure 1).

### Association of Fencing Response Following Concussion and Game-Day Clinical Outcomes

Group comparisons of game-day SAC neurological examination measures and game-day mBESS total scores are detailed in Figure 2. Descriptive statistics for these outcomes is presented in Table 2 (median scores reported below). There was no statistically

**TABLE 1. Demographics**

| Variable | Overall (n = 169) | Fencing (n = 57) | No fencing (n = 112) |
|---|---|---|---|
| Age, M ± SD | 25.8 ± 2.8 | 25.7 ± 2.7 | 25.8 ± 2.8 |
| Regular season (vs preseason), n (%) | 141 (83.4%) | 48 (84.2%) | 93 (83.0%) |
| Number of prior concussions, n (%) | | | |
| 0 | 101 (59.8%) | 33 (57.9%) | 68 (60.7%) |
| 1 | 29 (17.2%) | 10 (17.5%) | 19 (17.0%) |
| 2 | 23 (13.6%) | 5 (8.8%) | 18 (16.1%) |
| 3 | 12 (7.1%) | 7 (12.3%) | 5 (4.5%) |
| 4 | 3 (1.8%) | 1 (1.8%) | 2 (1.8%) |
| 5 | 1 (0.6%) | 1 (1.8%) | 0 (0.0%) |
| Position, n (%) | | | |
| Defensive back | 57 (33.7%) | 19 (33.3%) | 38 (33.9%) |
| Wide receiver | 49 (29.0%) | 17 (29.8%) | 32 (28.6%) |
| Tight end | 21 (12.4%) | 7 (12.3%) | 14 (12.5%) |
| Running back | 9 (5.3%) | 3 (5.3%) | 6 (5.4%) |
| Quarterback | 15 (8.9%) | 5 (8.8%) | 10 (8.9%) |
| Linebacker | 15 (8.9%) | 5 (8.8%) | 10 (8.9%) |
| Defensive line | 3 (1.8%) | 1 (1.8%) | 2 (1.8%) |

M, mean SD, standard deviation.

**TABLE 2. Descriptive Statistics Outcome Variables Between Players Who Exhibited a Fencing Response after Concussion to Players Who Did Not Exhibit a Fencing Response after Concussion**

| Variable | Fencing response | n | No fencing response | n |
|---|---|---|---|---|
| Days until return to full participation | 10.0 (5.5, 16.0) | 57 | 9.0 (5.3, 16.0) | 112 |
| Day-of-injury SAC | | | | |
| Immediate memory | 18.0 (15.0, 20.0) | 38 | 18.0 (15.0, 21.0) | 89 |
| Orientation | 5.0 (4.0, 5.0) | 38 | 5.0 (4.0, 5.0) | 89 |
| Concentration | 4.0 (3.0, 5.0) | 38 | 4.0 (3.0, 5.0) | 89 |
| Delayed recall | 3.0 (0.8, 5.3) | 38 | 5.0 (3.0, 7.0) | 89 |
| Total score | 27.5 (24.8, 34.0) | 38 | 30.0 (26.0, 35.5) | 89 |
| Day-of-injury mBESS | | | | |
| Total score | 4.0 (1.0, 5.5) | 37 | 4.0 (1.3, 7.0) | 84 |
| SCAT-5 Total Symptom Scores | | | | |
| Day-of-injury | 2.0 (0.0, 9.75) | 38 | 5.0 (1.0, 15.0) | 89 |
| Postinjury day 1 | 2.5 (0.0, 22.5) | 28 | 2.0 (0.0, 10.3) | 66 |
| Postinjury day 2 | 1.0 (0.0, 8.0) | 33 | 1.0 (0.0, 7.3) | 78 |
| Postinjury day 3 | 1.5 (0.0, 8.0) | 30 | 0.0 (0.0, 3.0) | 76 |
| Postinjury day 4 | 0.0 (0.0, 1.0) | 29 | 0.0 (0.0, 2.5) | 73 |
| Postinjury day 5 | 0.0 (0.0, 1.8) | 28 | 0.0 (0.0, 2.0) | 62 |
| Postinjury day 6 | 0.0 (0.0, 2.3) | 18 | 0.0 (0.0, 5.0) | 31 |
| Postinjury day 7 | 0.0 (0.0, 1.0) | 9 | 0.0 (0.0, 2.0) | 23 |
| Postinjury day 8 | 0.0 (0.0, 0.8) | 16 | 0.0 (0.0, 1.0) | 33 |
| Postinjury day 9 | 0.0 (0.0, 1.5) | 14 | 0.0 (0.0, 3.0) | 23 |
| Postinjury day 10 | 0.0 (0.0, 0.0) | 15 | 0.0 (0.0, 3.0) | 30 |

mBESS, modified Balance Error Scoring System; SAC, Standardized Assessment of Concussion; SCAT-5, Sport Concussion Assessment tool 5.
Values represent medians and interquartile ranges.

significant difference between groups in SAC immediate memory ($\chi^2$ = 0.03; fencing: 18, nonfencing: 18; $P$ = .86), SAC orientation ($\chi^2$ = 0.00; fencing: 5, nonfencing: 5; $P$ = .99), SAC concentration ($\chi^2$ = 0.45; fencing: 4, nonfencing: 4; $P$ = .50), SAC delayed recall ($\chi^2$ = 1.80; fencing: 5, nonfencing: 3; $P$ = .18), SAC total score ($\chi^2$ = 0.99; fencing: 27.5, nonfencing: 30; $P$ = .32), or mBESS score ($\chi^2$ = 0.03; fencing: 4, nonfencing: 4; $P$ = .87).

### Association of Fencing Response With Subjective Symptom Scores

Subjective assessment of concussion severity was assessed through the SCAT5 symptom scale scores. Group-level differences in symptom recovery based on fencing response status are shown in Figure 3. There was not a significant difference between groups in total symptom scores over time (β: −0.02; $P$ = .92).

## DISCUSSION

In this study of concussions in American professional football players across 6 NFL seasons, no clinically meaningful differences in cognitive/balance measures, symptom severity, or recovery were observed between players who exhibited a fencing response after concussion and those who did not exhibit a fencing response after concussion. Players sustaining a concussion with a fencing response received medical clearance (which included a separate neurocognitive evaluation by each Club's consulting neuropsychologist) to return to full participation in an equivalent time frame and had equivalent concussion severity on symptom recovery and objective examination findings on day of injury compared with a matched cohort of players with no documented fencing response after concussion. Wide receivers and defensive backs represented 2 in 3 cases of observed fencing response after

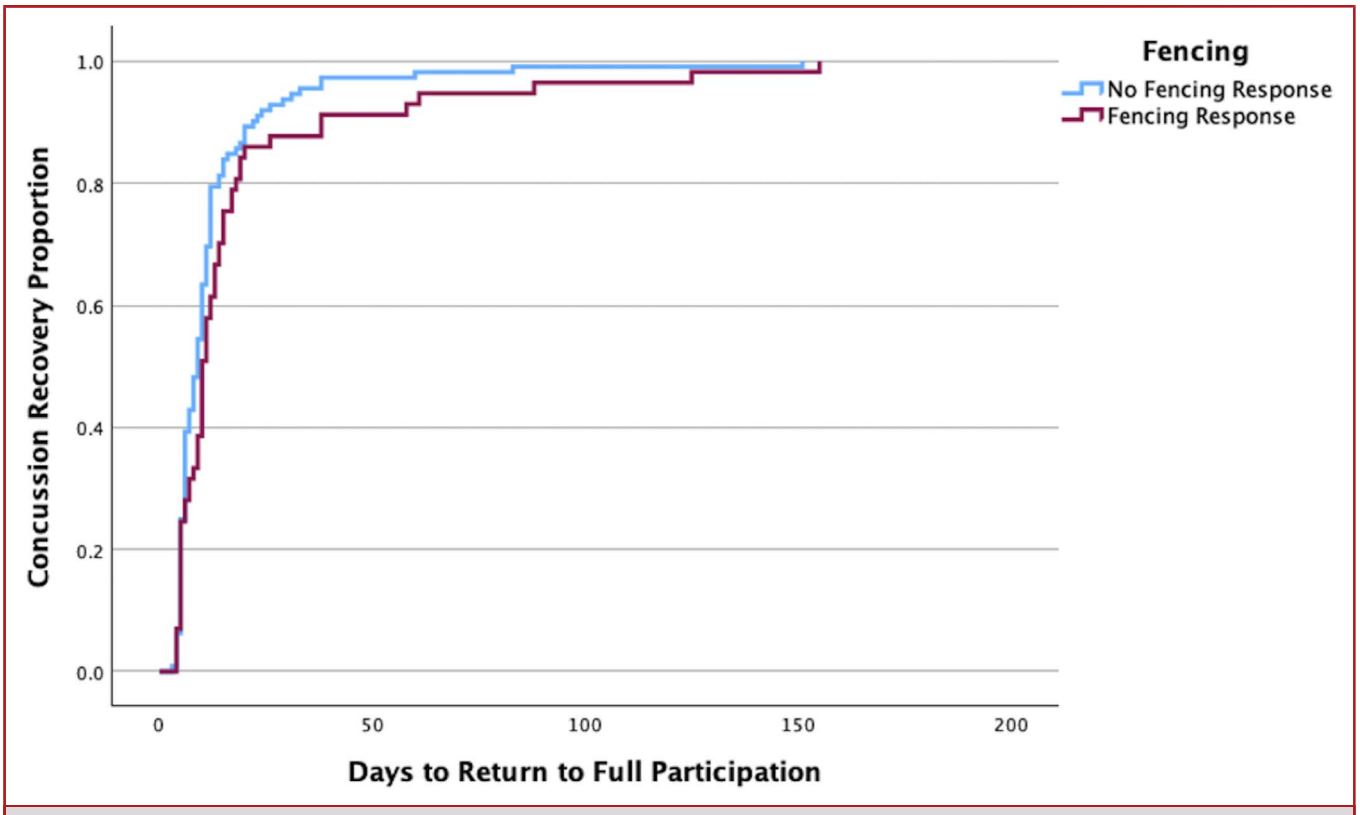


**FIGURE 1.** *Recovery proportion plot between players with fencing response after concussion (red line) to players without fencing response after concussion (blue line) derived from Cox regression analysis. No statistically significant difference in hazard for days until return to full participation was observed between groups (P = .34).*

concussion during in-game preseason and regular-season play in the study period.

Wide receivers and defensive backs are typically among the fastest runners on the field,[22,23] which could result in higher rotational head accelerations when engaging with another player.[24] Prior studies on tonic posturing are limited, but 1 animal study determined that tonic posturing was elicited at moderate (1.9 atmospheres of fluid pressure [atm]) but not mild (1.2 atm) head impact forces.[12] The authors hypothesized that tonic posturing is a neurological disturbance resulting from sufficient magnitude biomechanical forces on the brainstem leading to a vestibular reflex.[12] In human neurophysiology, activation of the lateral vestibular nucleus manifests as ipsilateral limb extension and flexor inhibition,[25] resulting in the appearance of a fencer.

Although greater velocities or magnitudes of force may be associated with the fencing response, force alone does not seem to determine whether a concussion occurs. There is significant heterogeneity among individuals regarding their response to head impact exposure, where small magnitude forces are sufficient to elicit concussion symptoms in some cases and insufficient in others.[26,27] In other words, fencing response may be an indicator of *biomechanical* severity but may not be an indicator of *clinical* severity of the injury. Based on the results of this study, a documented fencing response does not seem to be a significant mediator in objective examination findings, symptom severity, or return to participation in this population. However, the fencing response is a meaningful clinical sign that a concussion occurred,[11] and early detection of a concussion and removal from play is paramount to reducing the risk of prolonged recovery.[28] Our findings should not be interpreted that the absence of a fencing response means that a concussion has not occurred, but rather that observing a fencing response does not increase the risk of worse concussion outcomes.

## Limitations

This study has limitations worth noting. Days until return to full participation potentially could have been influenced by the team's schedule, such as an upcoming bye week after injury, and this was unable to be controlled for in our analyses. Data from additional recommended clinical outcome assessments, such as serial neurocognitive examinations or vestibular/ocular motor testing, were not available for this study. The SAC is a screening tool for cognitive performance with floor and ceiling effects,[29] it should be noted, however, that it is a requirement of the NFL's Head, Neck, and Spine Committee's Concussion Diagnosis and Management Protocol that all players sustaining a concussion undergo a formal neurocognitive assessment by that player's

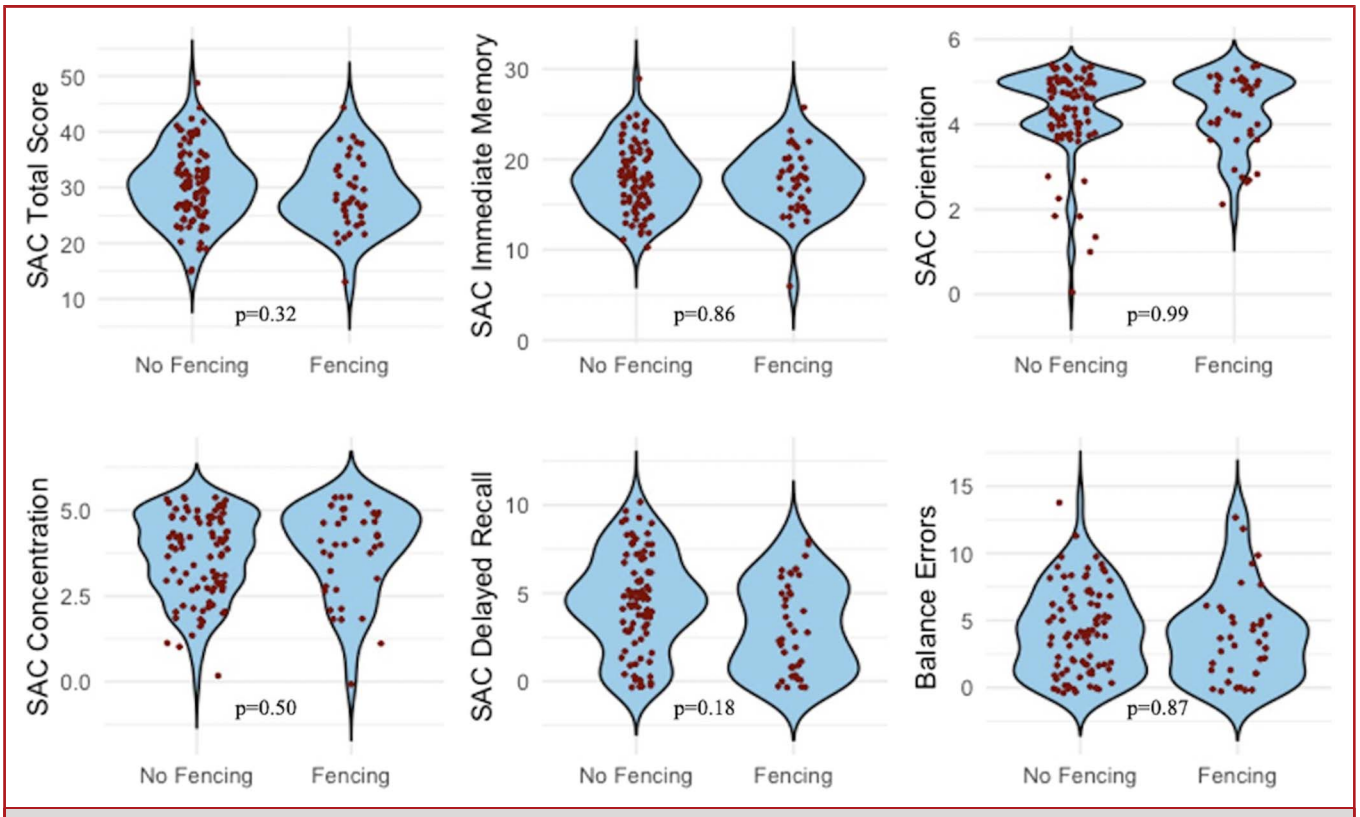


**FIGURE 2.** *Violin plot comparing game-day SAC and modified Balance Error Scoring System outcomes between players with fencing response after concussion and players without fencing response after concussion. Red dots depict individual data points, and the blue outline depicts the distribution of scores by groups. Alpha (p) for each group comparison is reported. SAC, Standard Assessment of Concussion.*

Club's consulting neuropsychologist and that the results of the assessment must be viewed as being back to baseline or, if a decrement is present, determined to be from a nonconcussion cause, before medical clearance to return from a concussion. Unfortunately, we were unable to access individual player neurocognitive test results for this study. The mBESS is a modified version of the original Balance Error Scoring System test created to expedite quick evaluations of concussion in sideline settings. The full BESS test may be more reliable and elicit more information on balance. In-depth concussion observations have been documented in the NFL using UNC reports since 2018, but this documentation is not available for concussions sustained during practices, which may have limited the available sample size. Designation of fencing response was based on observed, video-reviewed, and documented as fencing response by designated medical personnel; it is possible that a concussion with no documented fencing response had this response, but it was not documented in the UNC report. Data about whether a player had a history of fencing response or no fencing response were not available in this study. Mechanism of injury (ie, helmet to helmet contact, helmet to ground contact) was not made available for analysis or used for matching of participants. Linear/rotational forces related to concussion are not routinely measured in the NFL and were not available for this analysis. Furthermore, this study does not examine any potential long-term outcomes associated with the fencing response. These findings are only generalizable to professional American football players and should be formally studied in larger samples, other sports, and other populations (eg, youth sports, college athletics, etc.). NFL players play under a strict and highly-resourced medical protocol, so understanding concussion management across populations will be critical for future research.

## CONCLUSION

In this cohort study of professional American football players who sustained a concussion, there was no difference in day-of-injury neurological examination, symptom recovery, or days until return to full participation between players who exhibited tonic posturing (ie, a fencing response) after concussion compared with players who did not. Players whose position is associated with more frequent higher velocity impacts (eg, wide receivers and defensive backs) represented >60% of the total number of recorded fencing responses in the NFL from 2018 to 2023. From

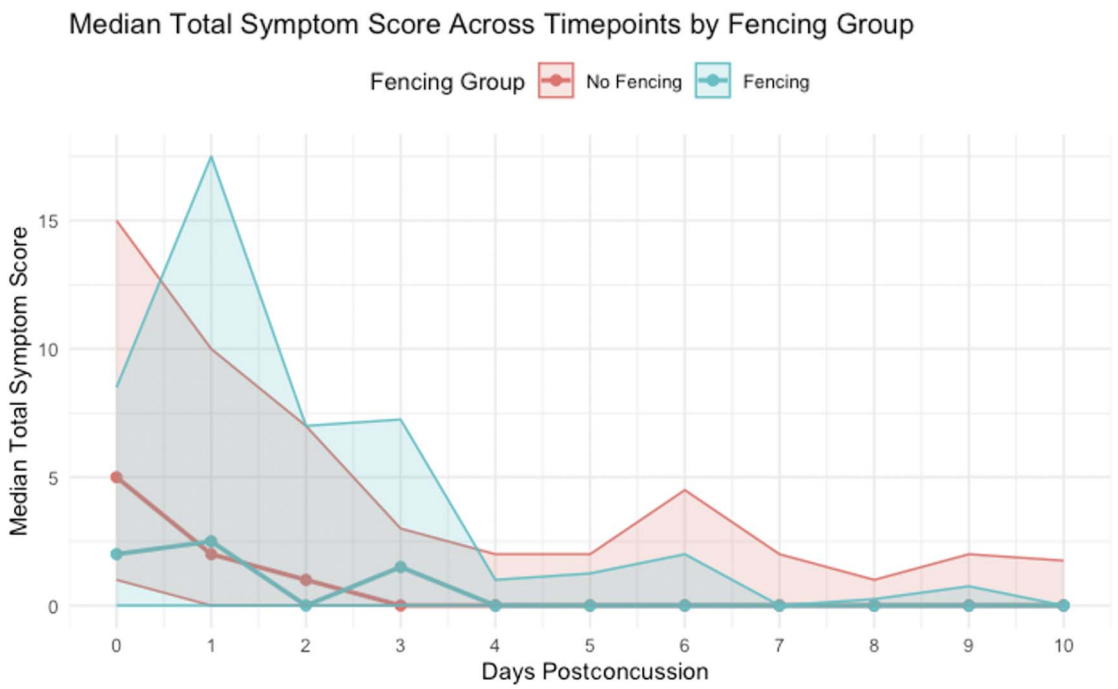


**FIGURE 3.** *Comparison of total symptom scores over time between players with a fencing response after concussion to players without a fencing response after concussion. The shaded area indicates the interquartile range whereas the dot indicates the group median on that day. The first data point is the median total symptom score on the day of injury (Day 0).*

a symptom-based, neurological examination, and recovery time perspective, fencing response after concussion was not associated with concussion severity or concussion recovery in professional American style football. Future research studies examining the fencing response after concussion—both acutely and from a longer-term perspective—are warranted.

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Allen Sills, Gary Solomon, and Douglas Terry are employees of the National Football League, whose data were used in this analysis. David O. Okonkwo is Team Neurosurgeon for the Pittsburgh Steelers Football Club and a member of the NFL Head, Neck, and Spine Committee. Erin B. Wasserman is an employee of IQVIA, which is in a paid consultancy with the NFL. Riju Shrestha is an employee of IQVIA, which is in a paid consultancy with the NFL. Douglas P. Terry receives funding from the National Football League, Amgen Inc; received funding from Football Research Inc; and is on the Scientific Advisory Board for HITIQ. Gary Solomon was a Senior Medical Advisor for the National Football League Player Health and Safety Department. The other authors have no personal, financial, or institutional interest in any of the drugs, materials, or devices described in this article.

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Author Contributions: DO, AM and SE conceived of the study and wrote the initial draft. SE ran the statistical analyses and created the figures. EW and RS completed the data extraction and organizing of the dataset. DT, GS, and AS provided operational support and consulted on study design. All authors critically reviewed the initial draft and approved the final version for submission.

# COMMENTS

The identification of clinical signs following head trauma or impact has become a crucial part of recognizing and managing concussion.[1a] The clinical signs, when observed from the sidelines or on video, allow for the immediate identification of concussion and removal from play.

The authors have examined one of the many clinical signs of concussion, "tonic posturing," which has been defined as an "involuntary sustained contraction of 1 or more limbs (typically upper limbs) so that the limb is held stiff despite the influence of gravity or the position of the player. The tonic posturing could involve other muscles such as the cervical, axial, and lower limb muscles."[1a] Tonic posturing, when present, is a clinical sign that is diagnostic of concussion. It has also been long established that tonic posturing itself is not an indicator of prolonged concussion recovery.[2a,3a]

The authors have reaffirmed the findings that tonic posturing does not relate to recovery time or concussion severity in a cohort of professional American football players and that this sign remains a valid diagnostic indicator of concussion. However, the authors' use of the term "fencing response" is inconsistent with modern internationally recognized classifications, including the International League Against Epilepsy (ILAE) classification of seizure types.[4a] Furthermore, the term does not align with international consensus definitions of video signs of concussion in professional sports,[1a] in which "tonic posturing" is the internationally accepted and validated terminology.

The term "fencing response" is used to describe the common presentation of tonic posturing, in which 1 arm is held in an extended position, similar to the *en garde* position of a fencing bout. However, it fails to acknowledge that tonic posturing can involve other muscles such as the cervical, axial, and lower limb muscles."[1a]

Adherence to international terminology and definitions eliminates confusion and helps advance the scientific process. Thus, we anticipate that future papers will exclusively incorporate modern and validated terminology, and that terms such as "fencing response" will be banished to the archives of antiquity.

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